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Drone Control System using ArduPilot

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Project Goal and Scope

- *To get familiar with the ArduPilot platform and its capabilities*
- *Set up the environment for drone testing*
- *Build and execute an autonomous flight mission*
- *Integrate components: GPS, WiFi, ESC, battery, flight controller*

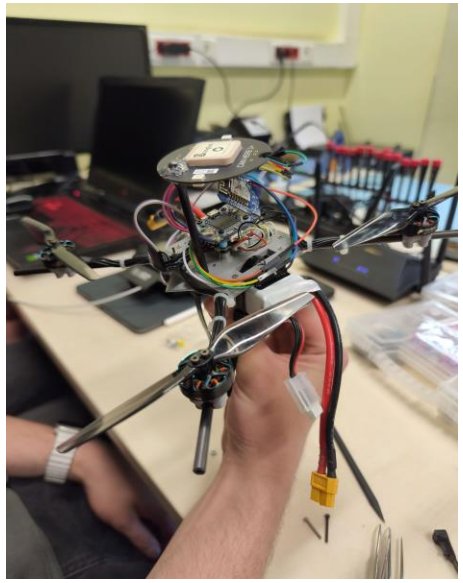
Methods and Tools

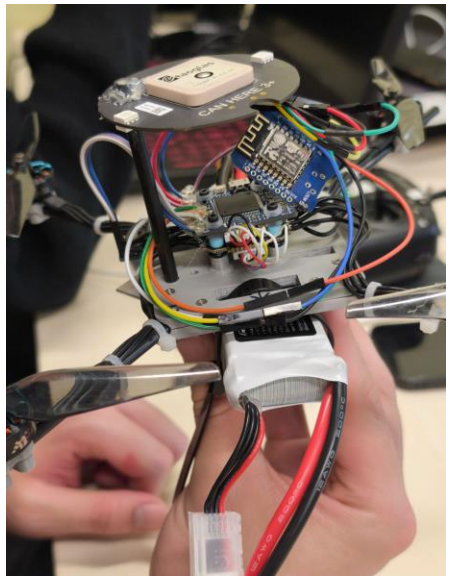
- *Operating system: Linux (installed and configured)*
- *Software: Mission Planner, ArduPilot*
- *Hardware: WiFi modules, GPS, ESC, battery, flight controller, drone frame*
- *Communication: WiFi router configured for remote connection*

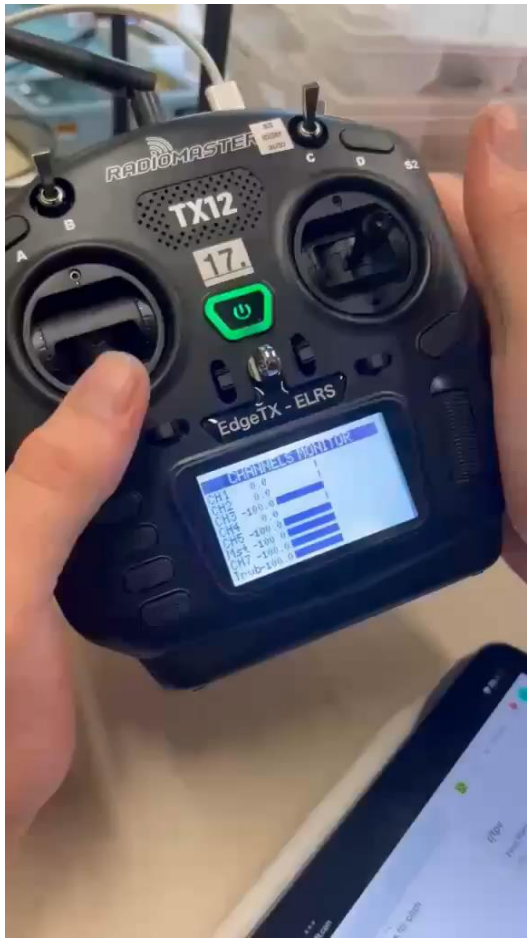


Project Stages

- *Installed Linux and configured ArduPilot*
- *Created a test mission using Mission Planner*
- *Configured WiFi modules and router*
- *Assembled the drone (GPS, WiFi, ESC, battery, flight controller, frame)*
- *Checked connectivity and configured parameters in ArduPilot*
- *Performed a successful short-duration flight test*



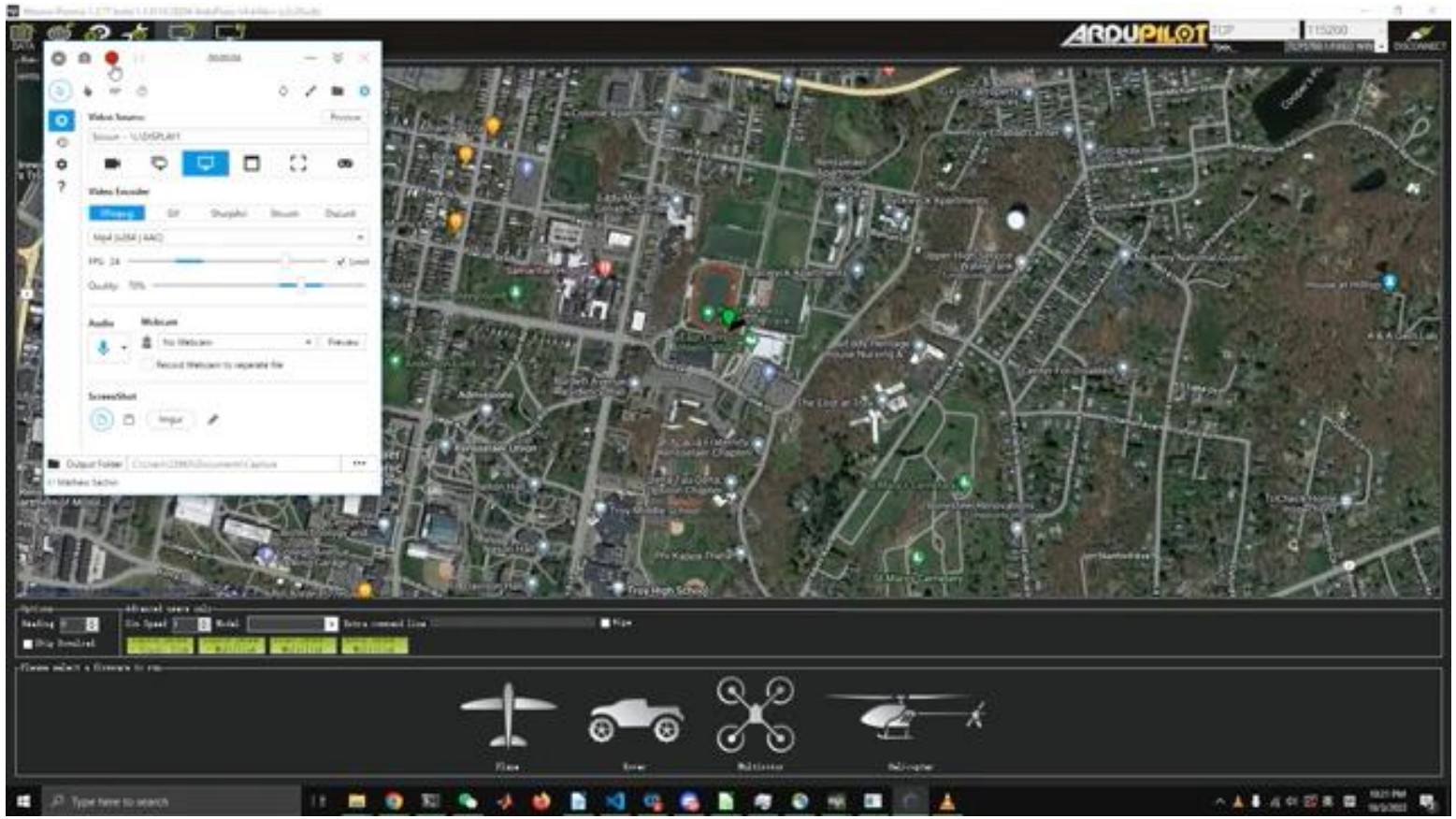




Demonstration

Mission Planner + SITL demo:

- *Takeoff → Waypoints → Landing*
- *Map view + path preview*
- *Real-time telemetry (altitude, speed)*



Results and Conclusions

- *The drone successfully took off and flew for several seconds*
- *We gained hands-on experience with embedded systems and wireless communication*
- *Combined knowledge from electronics, programming, and control systems*
- *The project could be extended with a camera or telemetry system*



Summary

- *The project covered the full process: software setup, configuration, and flight*
- *Provided practical skills in system integration and problem-solving*
- *We're happy with the result — a functional, flying drone*
- *Thank you for your attention — we’re happy to answer questions!*





Thank you for attention